

Asymptotics Basics

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Reference

1. Bruce E. Hansen (2023), "Econometrics", Chapter 6-7
2. Bruce E. Hansen (2022), "Probability and Statistics for Economists", Chapter 7-9

Convergence in probability

1. A sequence of real numbers a_n **converges** to a if for all $\epsilon > 0$, there exists N such that for all $n \geq N$,

$$|a_n - a| < \epsilon$$

We write $\lim_{n \rightarrow \infty} a_n = a$.

2. A sequence of random vectors $Z_n \in \mathbb{R}^k$ **converges in probability** to Z as $n \rightarrow \infty$, denoted $Z_n \rightarrow_p Z$, if for all $\delta > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\{\|Z_n - Z\| > \delta\} = 0$$

3. An estimator $\hat{\theta}$ of θ is **consistent** if $\hat{\theta} \rightarrow_p \theta$ as $n \rightarrow \infty$.

Convergence in probability

Example 1:

Suppose $Z_n \sim i.i.d \mathcal{N}(0, 1)$. Then, for any $\delta > 0$,

$$\mathbb{P}\{|Z_n - 0| > \delta\} = \mathbb{P}\{|Z_n| > \delta\} \not\rightarrow 0$$

as $n \rightarrow \infty$.

Example 2:

Suppose $Z_n = n^{-1}Z$ where $Z \sim \mathcal{N}(0, 1)$. Then, for any $\delta > 0$,

$$\mathbb{P}\{|Z_n - 0| > \delta\} = \mathbb{P}\{|Z| > n\delta\} \rightarrow 0$$

as $n \rightarrow \infty$.

Example 3:

Suppose $Z_n \sim i.i.d \mathcal{N}\left(0, \frac{1}{n}\right)$. Then, for any $\delta > 0$,

$$\mathbb{P}\{|Z_n - 0| > \delta\} = \mathbb{P}\{|\sqrt{n}Z_n| > \sqrt{n}\delta\} \rightarrow 0$$

as $n \rightarrow \infty$.

Convergence in probability

Suppose $f : \mathbb{R}^k \rightarrow \mathbb{R}^m$ is a continuous function and $Z_n \rightarrow c \in \mathbb{R}^k$. Then, we have¹

$$f(Z_n) \rightarrow_p f(c)$$

As a special case, let $Z_n \rightarrow_p c \in \mathbb{R}^k$ and $Y_n \rightarrow_p d \in \mathbb{R}$. Then, we have

$$Z_n Y_n \rightarrow_p cd$$

Note that $W_n = (Z_n', Y_n)'$ and $f : \mathbb{R}^{k+1} \rightarrow \mathbb{R}^k$ defined as

$$f(W_n) = Z_n Y_n$$

is continuous.

¹We are not going to prove this.

Markov Inequality

For any $\delta > 0$, we have

$$\mathbb{P}\{|X| > \delta\} \leq \delta^{-1} \mathbb{E}|X|$$

Proof.

Let $f(x)$ be pdf of X . Then,

$$\begin{aligned}\mathbb{P}\{|X| > \delta\} &= \int_{\{x: |x| > \delta\}} f(x) dx \\ &\leq \int_{\{x: |x| > \delta\}} \frac{|x|}{\delta} f(x) dx \\ &\leq \int \frac{|x|}{\delta} f(x) dx \\ &\leq \delta^{-1} \mathbb{E}|X|\end{aligned}$$

□

Corollary: $\mathbb{P}\{|X| > \delta\} = \mathbb{P}\{|X|^r > \delta^r\} \leq \delta^{-r} \mathbb{E}|X|^r$ for any $r > 0$.

Weak Law of Large Numbers

If $Y_i \in \mathbb{R}$ are i.i.d. and $\text{var}(Y) < \infty$, then as $n \rightarrow \infty$, we have

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \rightarrow_p \mathbb{E}Y$$

Proof.

For any $\delta > 0$, we have

$$\begin{aligned} \mathbb{P} \left\{ \left| \frac{1}{n} \sum_{i=1}^n Y_i - \mathbb{E}Y \right| > \delta \right\} &\leq \delta^{-2} \mathbb{E} \left| \frac{1}{n} \sum_{i=1}^n Y_i - \mathbb{E}Y \right|^2 = \delta^{-2} \mathbb{E} \left| \frac{1}{n} \sum_{i=1}^n (Y_i - \mathbb{E}Y_i) \right|^2 \\ &= \delta^{-2} \frac{1}{n^2} \mathbb{E} \left| \sum_{i=1}^n (Y_i - \mathbb{E}Y_i) \right|^2 = \delta^{-2} \frac{1}{n^2} \sum_{i=1}^n \mathbb{E} |(Y_i - \mathbb{E}Y_i)|^2 \\ &= \delta^{-2} \frac{1}{n} \mathbb{E} |(Y_i - \mathbb{E}Y_i)|^2 = \delta^{-2} \frac{1}{n} \text{var}(Y_i) \rightarrow 0 \end{aligned}$$



Example

Let $X_i \in \mathbb{R}^k, i = 1, \dots, n$ be i.i.d. random vectors. Let $\mathbf{X} \in \mathbb{R}^{n \times k}$ be a matrix collecting each observations in its rows. Then, we have

$$\frac{1}{n} \mathbf{X}' \mathbf{X} = \frac{1}{n} \sum_{i=1}^n X_i X_i' \rightarrow_p \mathbb{E} X_i X_i'$$

as $n \rightarrow \infty$. Suppose e_i satisfies $\mathbb{E} X_i e_i = 0$. Let $\mathbf{e} \in \mathbb{R}^{n \times 1}$ be a vector of observations. Then, we also have

$$\frac{1}{n} \mathbf{X}' \mathbf{e} = \frac{1}{n} \sum_{i=1}^n X_i e_i \rightarrow_p \mathbb{E} X_i e_i = 0$$

Example

Let $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ be the sample linear projection coefficient. Suppose

1. $(X_i, Y_i), i = 1, \dots, n$ are i.i.d.
2. (X, Y) satisfies²

$$Y = X'\beta + e$$

$$\mathbb{E}[Xe] = 0$$

Then, we have

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e}) \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}\end{aligned}$$

²This does not imply that the conditional expectation function $m(x)$ is linear in x .

Example

Thus,

$$\hat{\beta} - \beta = \left(\frac{1}{n} \mathbf{X}' \mathbf{X} \right)^{-1} \frac{1}{n} \mathbf{X}' \mathbf{e} \rightarrow_p 0$$

because $\left(\frac{1}{n} \mathbf{X}' \mathbf{X} \right)^{-1} \rightarrow_p (\mathbb{E} X_i X_i')^{-1}$ and $\frac{1}{n} \mathbf{X}' \mathbf{e} \rightarrow_p 0$. In other words, $\hat{\beta}$ is consistent.

We need some regularity conditions for these results to hold. Sufficient conditions are

1. $\mathbb{E} X_i X_i'$ invertible.
2. $\mathbb{E} \|X_i\|^4 < \infty$
3. $\mathbb{E} e_i^4 < \infty$

Convergence in distribution

Definition:

Let Z_n be a sequence of random vectors with distributions $F_n(u) = \mathbb{P}\{Z_n \leq u\}$. We say Z_n **converges in distribution** to Z as $n \rightarrow \infty$, denoted $Z_n \rightarrow_d Z$, if for all u at which $F(u) = \mathbb{P}\{Z \leq u\}$ is continuous, $F_n(u) \rightarrow F(u)$ as $n \rightarrow \infty$. We call $F(u)$ the asymptotic (limit) distribution of Z_n .

Convergence in distribution

Let $Z_n \rightarrow_d Z \in \mathbb{R}^k$ and $f : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be continuous. Then, we have

$$f(Z_n) \rightarrow_d f(Z)$$

as $n \rightarrow \infty$. This is called the **continuous mapping theorem**. Another property that we will use (without proof) is stated in the following. Let $Z_n \rightarrow_d Z$ and $Y_n \rightarrow_p c$. Then,

$$Z_n Y_n \rightarrow_d cZ$$

as $n \rightarrow \infty$.

Central limit theorem

Univariate CLT:

If $X_i \in \mathbb{R}$ are i.i.d., and $\text{var}(X_i) < \infty$, then

$$n^{-1/2} \sum_{i=1}^n (X_i - \mathbb{E}X_i) \rightarrow_d \mathbb{N}(0, \text{var}(X_i))$$

as $n \rightarrow \infty$.

Multivariate CLT:

If $X_i \in \mathbb{R}^k$ are i.i.d., and $\mathbb{E}\|X_i\|^2 < \infty$, then

$$n^{-1/2} \sum_{i=1}^n (X_i - \mathbb{E}X_i) \rightarrow_d \mathbb{N}(0, \Sigma_X)$$

where $\Sigma_X = \mathbb{E}(X_i - \mathbb{E}X_i)(X_i - \mathbb{E}X_i)'$.

Central limit theorem

Note that

$$\begin{aligned}\text{var} \left(n^{-1/2} \sum_{i=1}^n (X_i - \mathbb{E}X_i) \right) &= n^{-1} \text{var} \left(\sum_{i=1}^n (X_i - \mathbb{E}X_i) \right) \\ &= n^{-1} \sum_{i=1}^n \text{var} ((X_i - \mathbb{E}X_i)) \\ &= \text{var} ((X_i - \mathbb{E}X_i)) \\ &= \mathbb{E}(X_i - \mathbb{E}X_i)(X_i - \mathbb{E}X_i)'\end{aligned}$$

where in the second equality, we've used independence, in the third, we've used identically distributed. If we don't have i.i.d., the most general form of the limit variance would be given by

$$\lim_{n \rightarrow \infty} \text{var} \left(n^{-1/2} \sum_{i=1}^n (X_i - \mathbb{E}X_i) \right)$$

assuming it exists (is finite). This is called the **long-run variance** of X_i .

Example

Let $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ be the sample linear projection coefficient. Suppose

1. $(X_i, Y_i), i = 1, \dots, n$ are iid.
2. (X, Y) satisfies³

$$Y = X'\beta + e$$

$$\mathbb{E}[Xe] = 0$$

Then, we have

$$\begin{aligned}\hat{\beta} &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} \\ \Rightarrow \sqrt{n}(\hat{\beta} - \beta) &= \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1} \frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{e}\end{aligned}$$

³This does not imply that the conditional expectation function $m(x)$ is linear in x .

Example

Note that

$$\begin{aligned}\frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{e} &= \frac{1}{\sqrt{n}}\sum_{i=1}^n X_i e_i \\ &= \frac{1}{\sqrt{n}}\sum_{i=1}^n (X_i e_i - \mathbb{E}X_i e_i) \\ &\rightarrow_d \mathbb{N}(0, \Omega)\end{aligned}$$

where $\Omega = \mathbb{E}[X_i X_i' e_i^2]$. Thus,

$$\sqrt{n}(\hat{\beta} - \beta) = \underbrace{\left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}}_{\rightarrow_p (\mathbb{E}X_i X_i')^{-1}} \underbrace{\frac{1}{\sqrt{n}}\mathbf{X}'\mathbf{e}}_{\rightarrow_d \mathbb{N}(0, \Omega)}$$

which implies

$$\sqrt{n}(\hat{\beta} - \beta) \rightarrow_d \mathbb{N}(0, Q^{-1}\Omega Q^{-1})$$

where $Q = \mathbb{E}X_i X_i'$.

Example

Furthurmore, we have

$$\sqrt{n}(\hat{\beta} - \beta) \rightarrow_d \mathbb{N}(0, \sigma^2(\mathbb{E}X_iX_i')^{-1})$$

under the added assumption of conditionally homoskedasticity, i.e.,

$$\mathbb{E}[e_i^2|X_i] = \sigma^2(X_i) = \sigma^2$$

is independent of X_i . If we do not have conditional homoskedasticity (or if we have serial correlation in $\{X_ie_i\}$), we need to estimate the limit variance of $\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_ie_i - \mathbb{E}X_ie_i)$, i.e.,

$$\lim_{n \rightarrow \infty} \text{var} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_ie_i - \mathbb{E}X_ie_i) \right) =: \hat{\Omega}$$

Then, after estimating $\hat{Q} = \mathbb{E}_n X_i X_i' = \frac{1}{n} \sum_{i=1}^n X_i X_i'$, we estimate the variance of the limit distribution of $\sqrt{n}(\hat{\beta} - \beta)$ as

$$\hat{Q}^{-1} \hat{\Omega} \hat{Q}^{-1}$$

One can do this using the Newey-West (1987) estimator.